Year 13 Further Maths - Further calculus techniques

Similar questions - Inverse trigonometric and partial fraction integrals

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Given $y = \tan^{-1}\sqrt{3x}$, find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$. **[2]**

(b) Hence show that $\displaystyle\int_{1/3}^{1} \frac{1}{\sqrt{x}(1+3x)}\,\mathrm{d}x = k\pi$, where k is a number to be determined in exact form. **[4]**

2

(a) Show that $\dfrac{4}{1-x^4} \equiv \dfrac{2}{1-x^2} + \dfrac{2}{1+x^2}$. **[2]**

(b) Show that $\displaystyle\int_0^{1/\sqrt{3}} \frac{4}{1-x^4}\,\mathrm{d}x = \ln(a+\sqrt{b}) + \frac{\pi}{c}$ where a, b, c are integers to be determined. **[6]**

3

(a) In this question you must show detailed reasoning. Express $\dfrac{x^3+2x}{(x+1)(x^2+1)}$ in the form $A + \dfrac{B}{x+1} + \dfrac{Cx+D}{x^2+1}$. **[5]**

(b) Hence find $\displaystyle\int_0^2 \frac{x^3+2x}{(x+1)(x^2+1)}\,\mathrm{d}x$, giving your answer in the form $a + b\ln 3 + c\ln 5 + d\tan^{-1}2$. **[5]**

4

(a) Determine the constants A, B, C, D such that $\dfrac{x^2+27}{x^2(x^2+9)} \equiv \dfrac{A}{x} + \dfrac{B}{x^2} + \dfrac{Cx+D}{x^2+9}$. **[4]**

(b) In this part you must show detailed reasoning. Hence determine the exact value of $\displaystyle\int_3^{\infty} \frac{x^2+27}{x^2(x^2+9)}\,\mathrm{d}x$. **[6]**

5 Given that $y = \sin^{-1}(x^3)$, find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$. **[3]**

Question 1

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1}{1+3x}\cdot\dfrac{\sqrt{3}}{2\sqrt{x}} = \dfrac{\sqrt{3}}{2\sqrt{x}(1+3x)}$

(b) $\displaystyle\int \frac{1}{\sqrt{x}(1+3x)}\mathrm{d}x = \frac{2}{\sqrt{3}}\tan^{-1}\sqrt{3x}$. Between $\frac{1}{3}$ and 1: $\dfrac{2}{\sqrt{3}}\left(\frac{\pi}{3}-\frac{\pi}{4}\right) = \dfrac{2}{\sqrt{3}}\cdot\dfrac{\pi}{12} = \dfrac{\pi}{6\sqrt{3}} = \dfrac{\sqrt{3}}{18}\pi$. $k = \dfrac{\sqrt{3}}{18}$

Question 2

(a) $\dfrac{2}{1-x^2}+\dfrac{2}{1+x^2} = \dfrac{2(1+x^2)+2(1-x^2)}{1-x^4} = \dfrac{4}{1-x^4}$

(b) $\displaystyle\int \frac{2}{1-x^2}\mathrm{d}x = \ln\frac{1+x}{1-x}$ and $\displaystyle\int \frac{2}{1+x^2}\mathrm{d}x = 2\tan^{-1}x$. At $x=\frac{1}{\sqrt{3}}$: $\ln\dfrac{\sqrt{3}+1}{\sqrt{3}-1} = \ln(2+\sqrt{3})$ and $2\tan^{-1}\frac{1}{\sqrt{3}} = \frac{\pi}{3}$. So $\ln(2+\sqrt{3})+\dfrac{\pi}{3}$: $a=2$, $b=3$, $c=3$

Question 3

(a) Dividing: $x^3+2x = (x^3+x^2+x+1)-x^2+x-1$, so the expression is $1+\dfrac{-x^2+x-1}{(x+1)(x^2+1)}$. $-x^2+x-1 \equiv A'(x^2+1)+(Bx+C)(x+1)$: $x=-1$ gives $A'=-\frac{3}{2}$; x^2: $B=\frac{1}{2}$; constants: $C=\frac{1}{2}$. So $1-\dfrac{3/2}{x+1}+\dfrac{\frac{1}{2}x+\frac{1}{2}}{x^2+1}$, with $A=1$, $B=-\frac{3}{2}$, $C=\frac{1}{2}$, $D=\frac{1}{2}$

(b) $\left[x-\frac{3}{2}\ln(x+1)+\frac{1}{4}\ln(x^2+1)+\frac{1}{2}\tan^{-1}x\right]_0^2 = 2-\frac{3}{2}\ln 3+\frac{1}{4}\ln 5+\frac{1}{2}\tan^{-1}2$

Question 4

(a) $x^2+27 \equiv Ax(x^2+9)+B(x^2+9)+(Cx+D)x^2$. x^3: $A+C=0$; x: $9A=0$, so $A=C=0$; constant: $9B=27$, $B=3$; x^2: $B+D=1$, $D=-2$

(b) $\displaystyle\int_3^\infty\left(\frac{3}{x^2}-\frac{2}{x^2+9}\right)\mathrm{d}x = \left[-\frac{3}{x}-\frac{2}{3}\tan^{-1}\frac{x}{3}\right]_3^\infty = \left(0-\frac{\pi}{3}\right)-\left(-1-\frac{\pi}{6}\right) = 1-\frac{\pi}{6}$

Question 5

$\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1}{\sqrt{1-(x^3)^2}}\cdot 3x^2 = \dfrac{3x^2}{\sqrt{1-x^6}}$